Fairness in AI-Assisted Emergency Response Systems

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Abstract

Evaluate whether AI-assisted emergency dispatch and triage systems produce different response outcomes across demographic groups for similar emergency scenarios.

Keywords

fairness in AI, machine learning, algorithmic bias, emergency response, evaluation

1 Introduction

Emergency response systems operate in high-stakes, time-critical settings where decisions about dispatch, prioritization, and transport can have serious consequences. As algorithmic decision support is increasingly incorporated into emergency dispatch and triage, concerns have emerged that these systems may produce unequal response outcomes across demographic or geographic groups.

This project examines whether emergency response processes that incorporate algorithmic decision support produce different response outcomes across demographic groups when emergency scenarios are similar in severity. Rather than focusing on aggregate disparities alone, we emphasize *conditional fairness*– whether comparable emergencies receive comparable responses across groups.

Prior work on fairness in machine learning highlights that fairness can be meaningfully evaluated through disparities in decision outcomes, even when models are embedded within broader sociotechnical systems or are not directly observable [Barocas et al. 2023; Mehrabi et al. 2022]. Because proprietary AI systems used in emergency dispatch and triage are often not fully transparent, this project adopts a proxy modeling approach, using observed data to approximate algorithmic decision-making. By grounding the analysis in controlled comparisons, this work aims to provide an evidence-based assessment of fairness in public-sector, AI-supported decision-making.

2 Proposed Topics

The main focus of this project is fairness in AI-assisted emergency dispatch and triage systems. Specifically, we investigate whether response outcomes, such as response time, dispatch priority, or transport decisions, differ between demographic groups after taking into account the severity of the emergency. The project is guided by the following research questions:

- When emergency incidents are similar in severity, do response outcomes differ between demographic groups?
- If differences are observed, are they consistent with established fairness concepts such as conditional parity?
- To what extent can observed differences be explained by operational factors versus potential algorithmic or systemic bias?

3 Data and Models

To ensure feasibility and reproducibility, this project uses public, city-level emergency response datasets rather than large national datasets that require substantial computational infrastructure. Candidate dataset(s) includes:

- NYC EMS Incident Dispatch Data

Variables include initial and final call types, initial and final severity level codes, dispatch and incident response times, travel time components, and indicators related to transport to hospital. Together, these fields support fine-grained comparisons of response outcomes across emergency scenarios with similar levels of severity, which is central to the project's focus on conditional fairness.

Because individual level demographic attributes are not available, the analysis will rely on clearly documented geographic variables such as borough, ZIP code, and community district. These identifiers can be linked to publicly available census data to approximate demographic composition at the neighborhood level. All analyses will be conducted at the area level, and limitations related to geographic aggregation, such as sensitivity to spatial scale, will be acknowledged and discussed.

4 Methodology

The project will follow a structured and transparent workflow. Relevant variables related to response timing, dispatch priority, transport decisions, and call characteristics will be selected and filtered to ensure comparability across incidents. Emergency incidents will be stratified into severity tiers using available call-type or priority indicators.

Baseline machine learning models (e.g., logistic regression or tree-based classifiers) will be trained to predict response outcomes from call characteristics and severity indicators, serving as proxies for algorithmic decision support in settings where underlying AI systems are not directly observable. Within each severity tier, observed outcomes and model predictions will be compared across geographic and census-linked demographic groups using summary statistics and visualizations.

Fairness will be evaluated through conditional comparisons, such as differences in outcome rates or error rates within severity levels, with results interpreted in light of operational constraints, assumptions, and data limitations.

5 Expected Findings

We expect to observe some raw variation in emergency response outcomes across geographic locations and or demographic groups.

We recognize that factors such as population density and infrastructure constraints may contribute to this variation; thus, the analysis will condition on these contextual factors to focus on fairness-related differences in response outcomes. A primary focus of this project is whether any of these differences persist after accounting for emergency severity. Differences may vary depending on the outcome measure examined, and the analysis is designed to identify where fairness considerations are most relevant rather than to attribute intent or assign fault.

6 Relevance to Fairness in AI

This analysis aligns closely with our DSCI 531 course themes by examining algorithm supported decision making in a high impact public sector domain. It applies formal fairness concepts to observational data and highlights trade offs between efficiency, consistency, and equity in AI assisted systems. By grounding the analysis in controlled comparisons and transparent metrics, the project contributes a careful and quantitative perspective on fairness in real world AI applications.

References

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